

Semester – III

Course Name: Data Structures and Algorithms-I

Course Code: BCA305A

L (Hr.)	T (Hr.)	Pr (Hr.)	Credits
4	0	0	4

Course Objectives:

1. To impart the basic concepts of data structures and algorithms.
2. To understand concepts about searching and sorting techniques.
3. To understand basic concepts about stacks, queues, lists, trees and graphs.
4. To understanding about writing algorithms and step by step approach in solving problems with the help of fundamental data structures.
5. To impart the basic concepts of algorithms implementation in optimized time.

Unit 1

Introduction and overview: Introduction, basic terminology, elementary data organization, data structures, data structures operations, Abstract data types(ADT), Algorithms: Complexity, Time-space tradeoff.

Preliminaries:Introduction, Mathematical notation and functions, Algorithmic notations, control structures, complexity of algorithms, other asymptotic notation for complexity of algorithm, Sub algorithms, variables, data types.

Unit 2

String processing: Introduction, Basic terminology, storing strings, character data type, string as ADT, String operations, word/text processing, Pattern matching algorithms.

Arrays, Records and Pointers: Introduction, linear arrays, arrays as ADT, representation of linear arrays in memory, traversing linear arrays, inserting and deleting, sorting , bubble sort, Searching, linear search. Binary search, multidimensional arrays, representations of polynomials using arrays, Dynamic memory management, records structure, Representation of records in memory, parallel arrays, matrices and sparse matrices.

Unit 3

Linked Lists: Introduction of linked list, representation of linked list in memory traversing a linked list, searching a linked list, memory allocation garbage collection, insertion into a linked list, deletion from a linked list ,header linked list, circular linked list, two way list(doubly linked lists) Buddy systems.

Stacks, queues, recursion: Introduction, stacks, array representation of stacks, linked representation of stacks, stack as ADT, Arithmetic expression , polish notation, application of stacks, recursion, tower of Hanoi, implementation of recursive procedures by stacks, queues, linked representation of queues, queues as ADT, circular queues , Dequeues, priority queues, application of queue.

Unit 4

Trees: Introduction , Binary trees, representing binary trees in memory, traversing binary trees, Traversal algorithm using stacks, Binary search tree, searching ,inserting and deleting in binary tree, balanced binary tree, AVL search tree, B-Tree (searching , insertion, deletion), B+ Tree, Red Black tree, Heap, heapsort, path length and Huffman code.

Graphs and their applications: Introduction, graph theory, sequential representation of graphs, adjacency matrix, path matrix, warshall algorithm, shortest path, linked representation of graph, operation on graphs, traversing a graph, posets, topological sorting, spanning tree .

Unit 5

Sorting and searching: Introduction, sorting, insertion sort, selection sort, merge sort, shell sort, radix sort, searching and data modification and hashing.

Course Outcomes (COs):

Upon successful completion of this subject students will able

CO1: Ability to analyse algorithms and a algorithm correctness.

CO2: Ability to implement various techniques of link list.

CO3: Ability to describe stack, queue with linked list operation.

CO4: Ability to have knowledge of tree and graphs concepts.

CO5: Ability to summarize searching and sorting techniques

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcome	Program Outcome											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		H							M			
CO2						M				H		
CO3				H				M				
CO4						M					H	
CO5	H				M							L

H = Highly Related; M = Medium; L = Low

Text Books

1. R. G. Dromey, “How to Solve it by Computer”, Second Edition, Prentice-Hall of India, 2002.

2. ReemaThereja,, "Data Structure using C" , Second Edition , Oxford University Press, 2014

Reference Books

1. M. A. Weiss, "Data Structures and Algorithm Analysis in C", Second Edition, Pearson Education Asia, 2002.
2. ISRD Group, "Data Structures using C", Fifth Edition Tata McGraw Hill, 2007
3. Richard F. Gilberg, Behrouz A. Forouzan, "Data Structures – A Pseudocode" Third Edition Prentice-Hall of India, 2004.